

Funky Moose Amp

(German version see [README_de.md](#))

A highly versatile, state-of-the-art Bass Amplifier Plugin, specifically designed to bring the funk! Built with modern JUCE frameworks and featuring a dynamic, interactive UI.

Features Overview

1. Tuner & Input Section

- **High-Precision Tuner:** Always visible at the top left. Instantly reacts to your input signal.
- **Input Stage:** Control the amount of signal hitting the amp head.

2. Amp & Tone Module

The heart of the Moose. A custom-modeled solid-state preamp with optional tube characteristics. * **Gain:** Drive the preamp section from crystal clean to gritty overdrive. * **Tone Stack (Bass / Mid / Treble):** Highly responsive analog-modeled EQs. * **Volume:** Master output volume of the preamp section. * **Tube Saturation (Toggle):** Engages an asymmetric tube-style saturation curve with dynamic harmonic generation. * **Slap Mode (Toggle):** A mid-scoop (500Hz) paired with a bass & treble boost (+5dB), specifically tailored for modern slap-bass tones. * **Low Cut (Toggle):** HPF at 40Hz to clean out subsonic rumble and maintain tight low-end. * **Auto Gain:** Automatically compensates overall volume changes when adjusting the EQ knobs.

3. Smart Compressor

An ultra-clean, transparent VCA-style compressor. * **Controls:** Input (Drive), Threshold, Makeup Gain, Attack, Release. * **Ratio Selection:** 4:1, 8:1, 12:1, and 20:1 (Limiting). * **Punch Circuit (Toggle):** Enhances the attack phase and tightens the low-end grip, making bass lines cut through any mix. * **Auto-Makeup (AUTO):** Analyzes gain reduction and automatically corrects output level. * **Dry/Wet Mix:** Built-in parallel compression without phase issues. * **Gain Reduction Meter:** Real-time visual feedback of compression amount.

4. Effects Section (ModFX)

Built-in high-quality effects, routeable in series or parallel. * **Octaver:** Sub-bass generator. Blends analog-style -1 Octave and +1 Octave signals. * **Envelope Filter (Auto-Wah):** Touch-sensitive filter. Controls: Attack, Decay, Range. * **Phaser:** Classic 4-stage optical phaser with Rate, Colour (Feedback), and Mix controls. * **Chorus:** Lush analog-style stereo bucket-brigade chorus. * **Parallel Mode (Toggle):** Move the Chorus & Phaser into a parallel signal path alongside the dry/amp signal to retain maximum low-end punch while adding width.

5. Master Section & Custom Cab IRs

- **Master Out:** Controls the final plugin output.
- **Mono Maker:** Sums all frequencies below a certain threshold to mono, ensuring phase-coherency in the crucial low registers.
- **Master Dry/Wet:** Mix the entire processed signal with your DI signal.
- **CAB IR Loader (CAB: OFF/ON):** Load custom Impulse Responses. Includes built-in custom "Funky Moose" Bass Cab responses. Easily switchable and mixable (IR Mix).

6. The "Funky Moose" Interactive UI

The Elch (Moose) is not just a logo; it's a dynamic visualizer of your tone! * **Level Visualization:** The Moose reacts to both your input and output RMS levels. * **Compression Feedback:** Heavy gain reduction affects the background texture and glow. * **Sunglasses Glow:** Engaging the "Punch" circuit on the compressor activates a bright neon glow on the Moose's sunglasses. * **State-of-the-Art Look & Feel:** Features a skeuomorphic aesthetic with realistic shadows, glowing toggle lamps (States), and a custom Pop-up preset system.

Presets & MIDI

- **Presets:** Includes an entire Factory Bank covering Clean DI tones, Vintage Motown, Modern Slap, Synth Bass, and Heavy Overdrives. Supports saving User Banks.
- **MIDI Learn:** Just click "MIDI LRN" and turn a knob on your physical controller (e.g., Akai MPK Mini) to map CC parameters instantly to any knob.

Build Instructions

Can be built as VST3, AU, and a Standalone App using CMake. `bash cmake -B build cmake --build build --target FUNKY_MOOSE_AMP_Standalone`